

The Model: Feature-Composed Item Factors

SAMPLE v2 (thesis-write Phase C, 2026-08-21) | discussion material, not thesis prose.
ProtoMF-paper symbols, LightFM-style composition; scope cut to delta-vs-host + grounding
mechanism per Matteo’s v1 feedback.

We introduce our approach as feature-composed I-ProtoMF, in short fl-ProtoMF. It differs from I-ProtoMF [2] in exactly one place. The host draws the embedding $\mathbf{t}$ of every item from a lookup table with one free learned row per item. fl-ProtoMF replaces this table by a composition in the manner of LightFM [1],

$$\mathbf{t} = \sum_{f \in \mathcal{F}_t} \mathbf{e}_f (+ \mathbf{e}_{\text{ID}(t)}), \quad (1)$$

where $\mathcal{F}_t$ holds the feature values recorded for the item, each $\mathbf{e}_f \in \mathbb{R}^d$ is a learned vector, and $\mathbf{e}_{\text{ID}(t)}$ is an optional per-item row discussed below. Everything downstream remains the host. The L^t item prototypes $\mathbf{p}_l^t$ stay free learned vectors, the item representation $\mathbf{t}^* \in \mathbb{R}^{L^t}$ is still the vector of similarities $\text{sim}(\mathbf{t}, \mathbf{p}_l^t)$, the user is still a free vector $\mathbf{u} \in \mathbb{R}^{L^t}$, and the score $\text{I-score}(u, t) = \mathbf{u}^\top \mathbf{t}^*$ is trained with the host’s loss and regularizers (Section 2.3). The model name keeps the f of feature rather than attribute because the composition sums whatever encoded rows it is given, the ID row being the first example, and choosing attribute-derived rows is what makes the resulting model interpretable.

The composition turns attribute values into learned words. Every distinct value that appears in the catalogue, the colour group black or the garment group jersey, owns exactly one row $\mathbf{e}_f$, shared by all items that carry it. An item is then spelled from these words. A plain black t-shirt sums the rows of its recorded values for department, product type, section, colour group, and pattern, and a second shirt that differs only in colour shares four of its five summands. The host’s item table, one independent row per item, becomes a small vocabulary of attribute rows from which all item embeddings are assembled. The rows are trained by the collaborative signal like any other parameter, but each row receives the gradients of every interaction with every item that carries its value, so what a row comes to encode is the collaborative meaning of one attribute value across the catalogue.

This is the grounding announced in the previous section, stated precisely. What the composition constrains directly is the item factor. Every $\mathbf{t}$ is a sum of rows that carry meaning on their own. The prototypes are not constrained at all. They remain free vectors, but they anchor a space that is now populated by attribute words, and a prototype can therefore be read by comparing it to those words directly in parameter space, with no detour through training data. Grounding the prototypes is in this sense shorthand for grounding the space they anchor, and the read-outs this buys

are developed in Section 5.4.

The optional ID row. The row $\mathbf{e}_{\text{ID}(t)}$ in Equation 1 is a free per-item vector added to the sum by configuration. It restores per-item capacity, letting an item deviate from what its attribute values alone would predict, at the price of an ungrounded summand in every embedding. It also makes the host a special case. With $\mathcal{F}_t$ empty and the ID row kept, the model reduces to I-ProtoMF exactly, and our implementation reproduces the host bit-identically in this configuration. The chapter therefore evaluates three arms. fl-ProtoMF uses features and the ID row, the noid arm uses features alone, and the f0 arm is the host reached through the extended code path, the reference point of all comparisons. What the ID row contributes, and what its ungrounded share costs, is measured in Sections 5.4 and 5.5.

References

- [1] M. Kula. Metadata embeddings for user and item cold-start recommendations. In *Proceedings of the 2nd Workshop on New Trends on Content-Based Recommender Systems (CBRecSys) at RecSys*, 2015.
- [2] A. B. Melchiorre, N. Rekabsaz, C. Ganhör, and M. Schedl. Protomf: Prototype-based matrix factorization for effective and explainable recommendations. In *Proceedings of the 16th ACM Conference on Recommender Systems (RecSys)*, pages 246–256, 2022.